Corso di SISTEMI OPERATIVI
Gruppi A e B

Prof. G. Ianni

Dott. D. Angilica

Sito del corso:

qui

Gruppo FB («Sistemi Operativi e Reti – UNICAL - INFORMATICA»):

https://bit.ly/2Er8HBQ

Microsoft Teams:
https://bit.ly/2ElmdXs

Microsoft Teams

Slide aggiornate all'A.A. 2025-26

SCAN ME

Iscrizione alle attività di laboratorio

Modalità d’esame e propedeuticità

- **Ci si può presentare all’esame** solo se si è sostenuto ed è regolarmente presente su ESSE3:
  «Fondamenti di programmazione»,
  «Architettura degli elaboratori» (Per chi ha il vecchio piano di studi)

NO «Me lo verbalizza domani», «L’avevo quasi passato»,
«Devono uscire i risultati», «Mio nonno è stato colpito da un meteorite»

- Bisogna avere **almeno** seguito «Programmazione e algoritmi» (ex «Fondamenti di programmazione 2»)
- **Fortemente richiesta la conoscenza di un linguaggio a oggetti (meglio se Python)**

- SISTEMI OPERATIVI:
  Prova scritta al calcolatore (con pre-test)

- SISTEMI OPERATIVI E RETI (corso disattivato): esame in due parti
  1. Scritto in laboratorio sul modulo di SISTEMI OPERATIVI (50% del voto). La prova è identica a quella di SISTEMI OPERATIVI e si svolge nelle stesse date
  2. Orale con discussione progetto sul modulo di RETI DI CALCOLATORI (50% del voto)

* ORARIO
* DIFFICOLTÀ DELL’ESAME
* RICEVIMENTO, DIRITTI E DOVERI DELLO STUDENTE
* GESTIONE DELLE RACCOMANDAZIONI

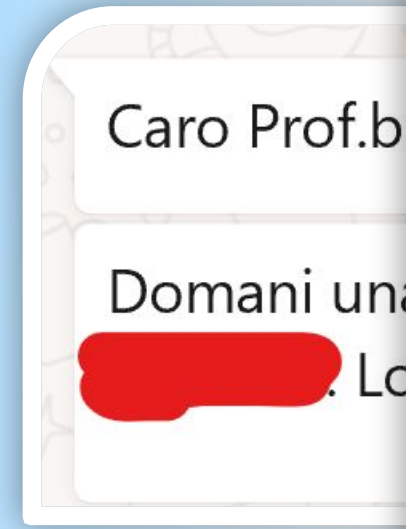
Fondamenti di programmazione

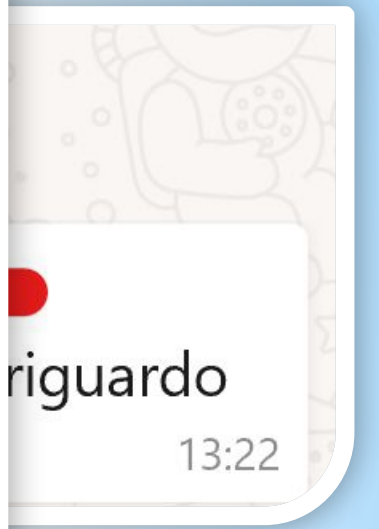
Sistemi operativi

Propedeuticità

Difficoltà dell’esame

"***Ognuno di noi è un genio. Ma se giudichi un pesce dalla sua abilità di scalare un albero, esso passerà il resto della sua vita a sentirsi uno stupido***"
(A. Einstein. Anzi no, Abramo Lincoln. Anzi no, Mark Twain. Anzi no, Winston Churchill. Anzi no. Marilyn Monroe)





Diritti e doveri dello studente
Gestione delle raccomandazioni

NEWS E FOCUS

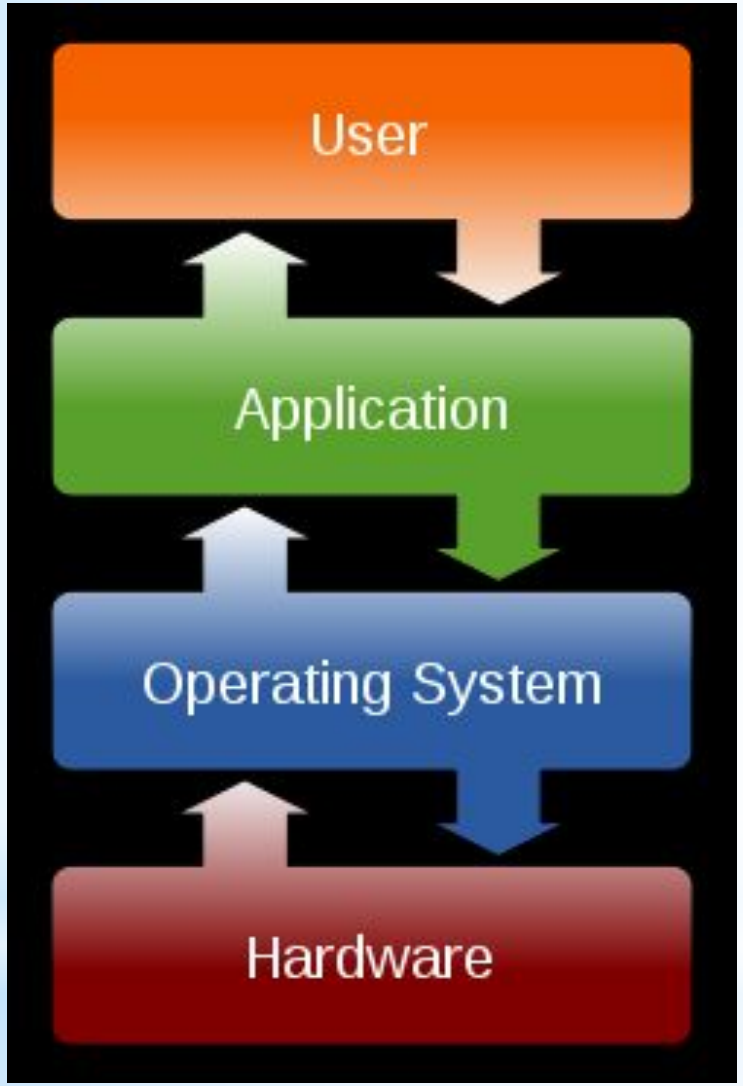
Chirurgo sbaglia e rimuove il fegato al posto della milza: paziente muore in Florida

Una donna denuncia il medico per negligenza dopo la tragica morte del marito in Florida. Il dottore avrebbe cercato di coprire l'errore

Programma del corso di Sistemi Operativi

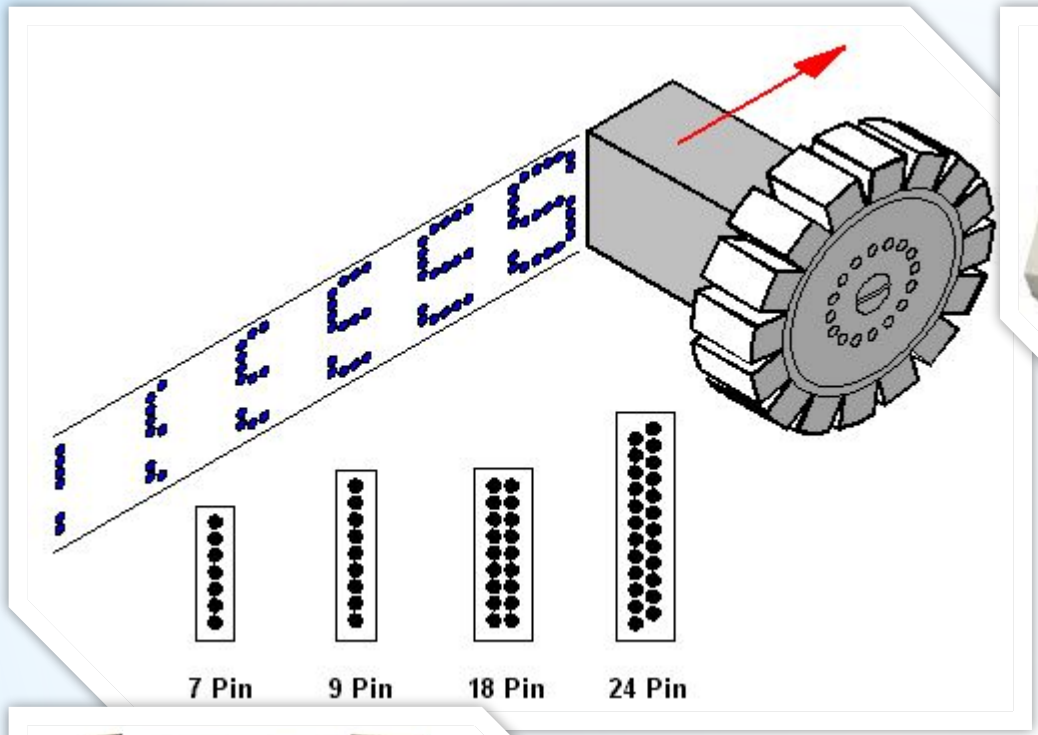
* **Programma del Modulo:**
  * Teoria dei sistemi operativi: gestione dei file, dei processi e della memoria.
  * Shell e linguaggi di scripting (bash + perl)
  * Programmazione multithreading (su Python)
  * Case studies: Linux, Windows, Sistemi Operativi Mobili

* **Testi suggeriti e materiale:**
  * **A. Silberschatz: Operating Systems Concepts** (9 o **10ma edizione**). Addison-Wesley,
  * A. Tanenbaum, Modern Operating Systems 5ed
  * Deitel & Deitel, Operating Systems.
  * Python 3 Concurrency: https://docs.python.org/3/library/concurrency.html
  * Learning Perl, ed. O'Reilly.
  * Materiale disponibile sul sito del corso
  * TIENI D'OCCHIO IL TEAM E LE RISORSE CHE METTIAMO SUL TEAM



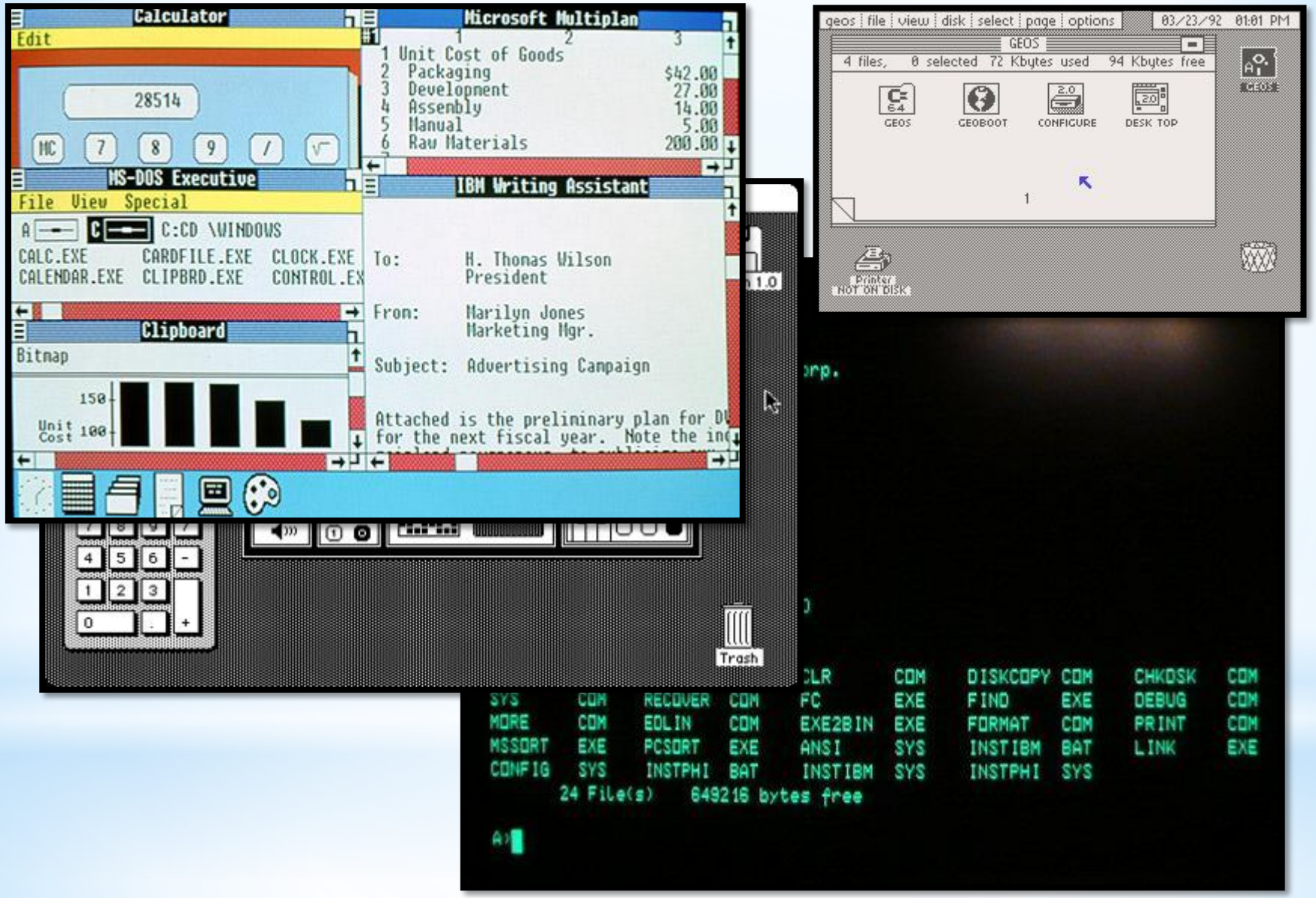
Visione astratta di un SO

*Tanto tempo fa sul pianeta Terra…

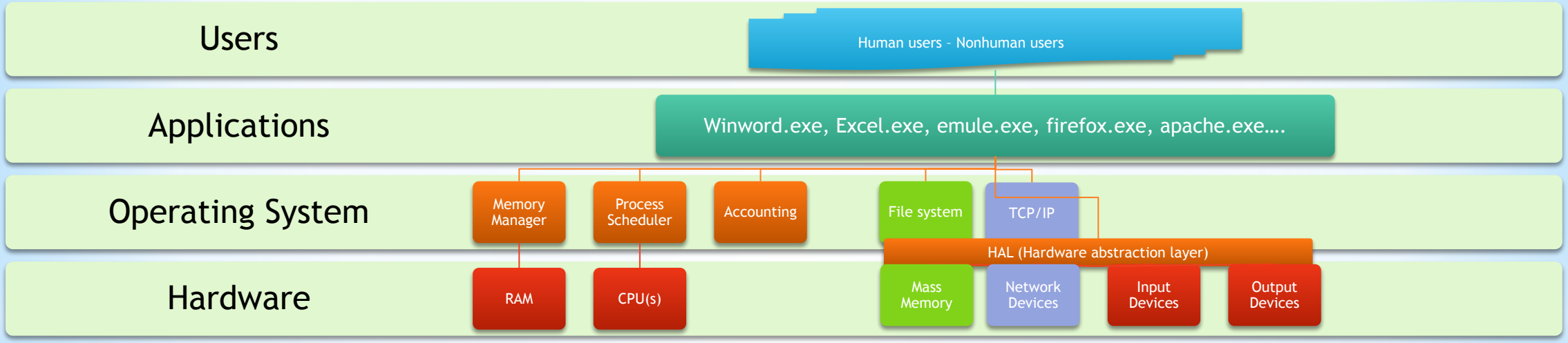


https://www.youtube.com/watch?v=Mopg8SMFuw0

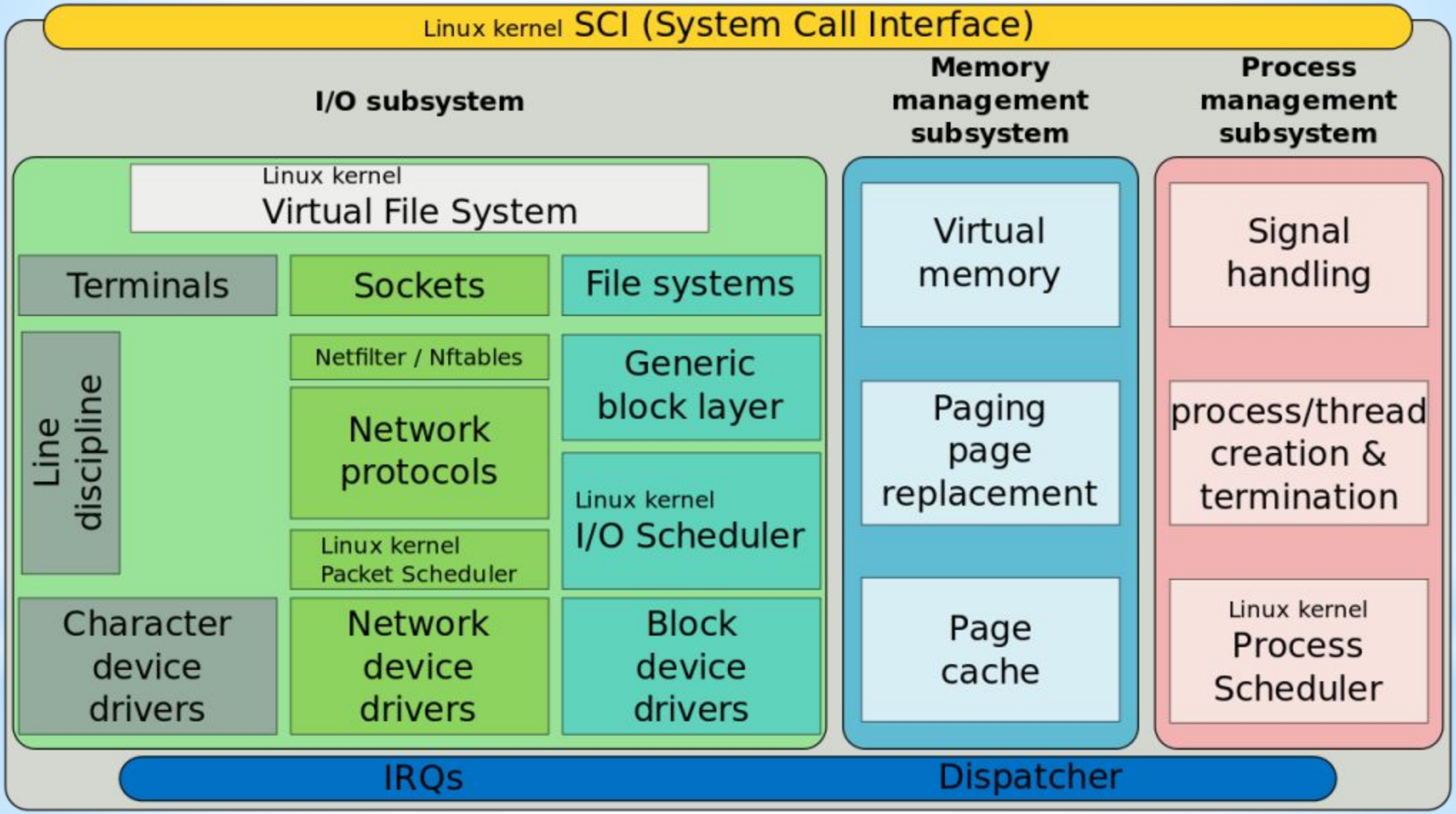
*Stampanti…



https://virtualconsoles.com/online-emulators/c64/



Visione di dettaglio


Linux kernel SCI (System Call Interface)
I/O subsystem
Memory management subsystem
Process management subsystem
Linux kernel
Virtual File System
Terminals
Sockets
File systems
Line discipline
Netfilter / Nftables
Network protocols
Generic block layer
Linux kernel
I/O Scheduler
Linux kernel
Packet Scheduler
Character device drivers
Network device drivers
Block device drivers
Virtual memory
Paging page replacement
Page cache
Signal handling
process/thread creation & termination
Linux kernel
Process Scheduler
IRQs
Dispatcher

Cos’è un sistema operativo

* Scopo primario: ***Gestire e astrarre*** *le* ***risorse*** *di un calcolatore*

* Il sistema operativo
  1. Astrae l’uso delle periferiche tramite un interfaccia comune
     - Funzioni di libreria comuni per le più grandi categorie di periferiche
  2. Gestisce la memoria primaria (RAM)
     - Allocazione e Protezione
  3. Gestisce la memoria secondaria (Dischi fissi, ecc.)
     - Mappatura sul dispositivo fisico
     - Organizzazione
     - Gestione degli errori
  4. **Gestisce i processi e i thread**
     - **Scheduling, accesso condiviso alle risorse**
  5. Gestisce gli eventi e l’interfaccia utente
     - Cattura e smistamento degli eventi ai processi
  6. Gestisce i diritti di accesso e la presenza di più utenti
     - Accesso multiplo, quote, diritti di accesso
  7. Gestisce la rete
     - Applicazioni distribuite, condivisione di risorse, socket

* Non più solo un gestore di CPU, RAM e disco
* Un sistema di isolamento e compartimentazione
* Una piattaforma per ambienti virtuali (WSL2, Hyper-V, VMware, KLM, Docker)
* Un mediatore tra hardware complesso e software (Multi-core, GPU, TPU…)
* Il primo muro fortificato a difesa del sistema (Controllo accessi, firewall, isolamento)

Cos’è un sistema operativo nel 2026

CreateFileA function (fileapi.h)

12/05/2018 • 29 minutes to read

Creates or opens a file or I/O device. The most commonly used I/O devices are as follows: file, file stream, di
volume, console buffer, tape drive, communications resource, mailslot, and pipe. The function returns a han
access the file or device for various types of I/O depending on the file or device and the flags and attributes

To perform this operation as a transacted operation, which results in a handle that can be used for transacte
CreateFileTransacted function.

Syntax

C++

```
HANDLE CreateFileA(
  LPCSTR                lpFileName,
  DWORD                 dwDesiredAccess,
  DWORD                 dwShareMode,
  LPSECURITY_ATTRIBUTES lpSecurityAttributes,
  DWORD                 dwCreationDisposition,
  DWORD                 dwFlagsAndAttributes,
  HANDLE                hTemplateFile
);
```

Parameters

`lpFileName`

The name of the file or device to be created or opened. You may use either forward slashes (/) or backslashe

In the ANSI version of this function, the name is limited to **MAX_PATH** characters. To extend this limit to 32,
the Unicode version of the function and prepend "\\?\" to the path. For more information, see Naming Files

For information on special device names, see Defining an MS-DOS Device Name.

```
OPEN(2)                Linux Programmer's Manual                OPEN(2)
```

NAME top

```
       open, openat, creat - open and possibly create a file
```

SYNOPSIS top

```
       #include <fcntl.h>

       int open(const char *pathname, int flags);
       int open(const char *pathname, int flags, mode_t mode);

       int creat(const char *pathname, mode_t mode);

       int openat(int dirfd, const char *pathname, int flags);
       int openat(int dirfd, const char *pathname, int flags, mode_t mode);

       /* Documented separately, in openat2(2): */
       int openat2(int dirfd, const char *pathname,
                   const struct open_how *how, size_t size);

   Feature Test Macro Requirements for glibc (see
   feature_test_macros(7)):

       openat():
           Since glibc 2.10:
               _POSIX_C_SOURCE >= 200809L
           Before glibc 2.10:
               _ATFILE_SOURCE
```

DESCRIPTION top

```
       The open() system call opens the file specified by pathname.  If
       the specified file does not exist, it may optionally (if O_CREAT
       is specified in flags) be created by open().
```

Funzioni di libreria

Process Monitor - Sysinternals: www.sysinternals.com

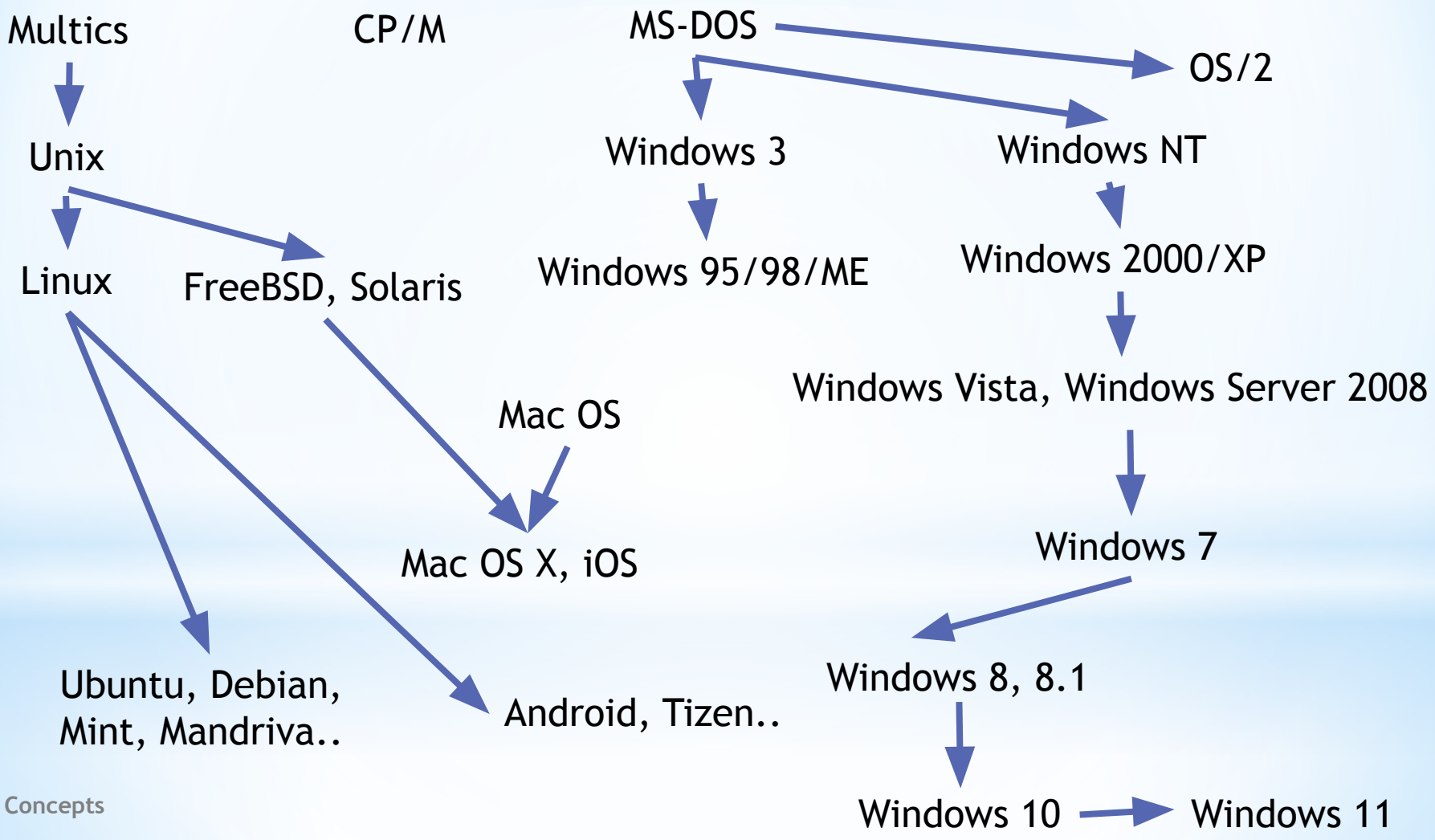
File Edit Event Filter Tools Options Help

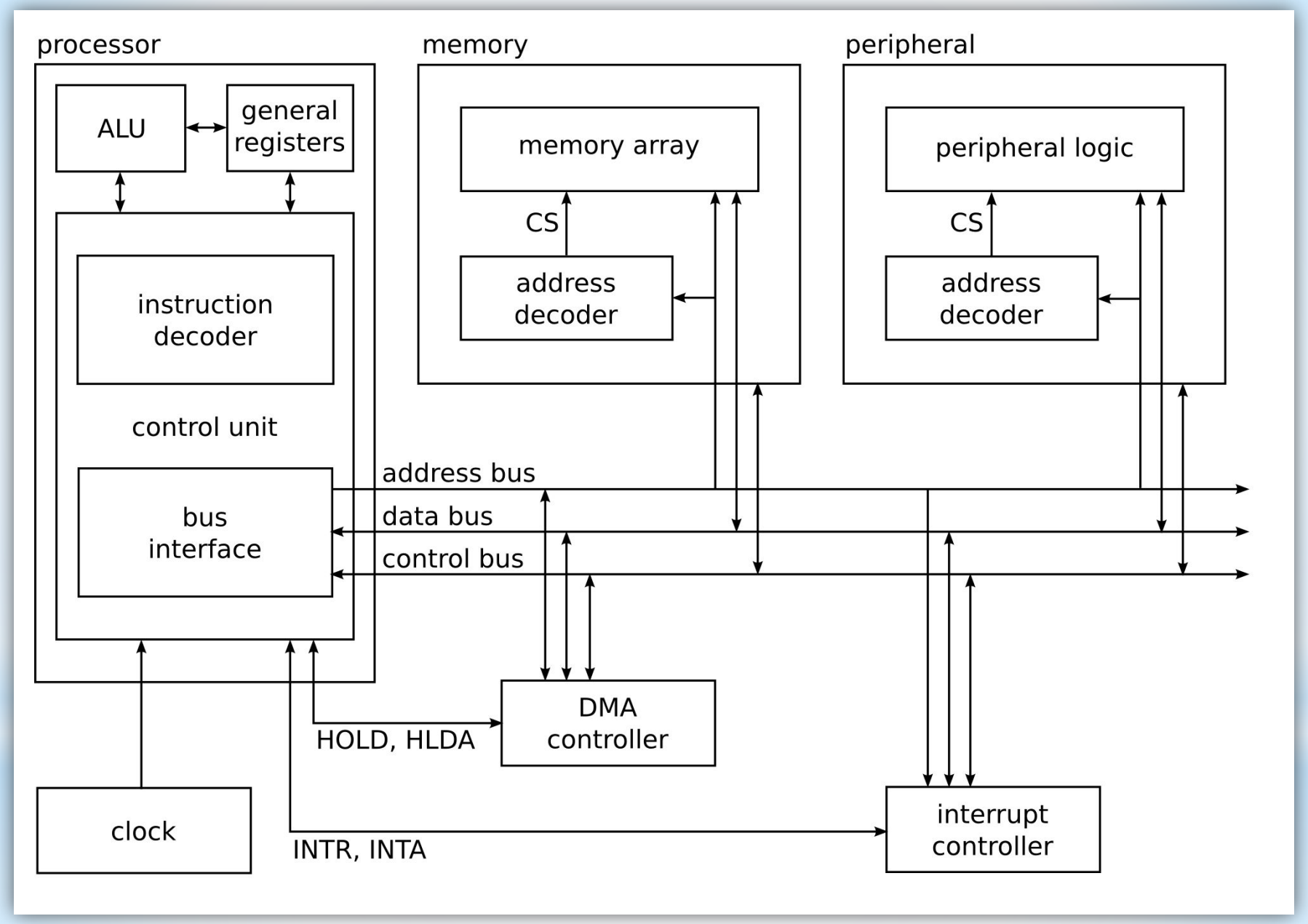
Time o...	Process Name	PID	Operation	Path
19:19:30...	Explorer.EXE	13396	IRP_MJ_READ	C:\Windows\System32\TaskFlowDataEngine.dll
19:19:30...	Procmon64.exe	195896	RegQueryValue	HKLM\System\CurrentControlSet\Control\WMI\Security\92ba8d4a-94ca-11ec-9b50-a87eeab25ed9
19:19:30...	Procmon64.exe	195896	RegQueryValue	HKLM\System\CurrentControlSet\Control\WMI\Security\9e814aad-3204-11d2-9a82-006008a86939
19:19:30...	System	4	Thread Create	
19:19:30...	Procmon64.exe	195896	Thread Create	
19:19:30...	Explorer.EXE	13396	RegOpenKey	HKCR\C:\Users\Ag
19:19:30...	Explorer.EXE	13396	RegQueryKey	HKCU\Software\Cl
19:19:30...	Explorer.EXE	13396	RegQueryKey	HKCU\Software\Cl
19:19:30...	Explorer.EXE	13396	RegQueryKey	HKCU\Software\Cl
19:19:30...	Explorer.EXE	13396	RegOpenKey	HKCU\Software\Cl
19:19:30...	Explorer.EXE	13396	RegOpenKey	HKCR\C:\Users\Ag
19:19:30...	Procmon64.exe	195896	Thread Create	
19:19:30...	Explorer.EXE	13396	RegQueryKey	HKCU\Software\Cl
19:19:30...	Explorer.EXE	13396	RegQueryKey	HKCU\Software\Cl
19:19:30...	Procmon64.exe	195896	RegOpenKey	HKLM\HARDWAR
19:19:30...	Explorer.EXE	13396	RegQueryKey	HKCU\Software\Cl
19:19:30...	Procmon64.exe	195896	RegQueryValue	HKLM\HARDWAR
19:19:30...	Explorer.EXE	13396	RegOpenKey	HKCU\Software\Cl
19:19:30...	Procmon64.exe	195896	RegCloseKey	HKLM\HARDWAR
19:19:30...	Explorer.EXE	13396	RegOpenKey	HKCR\C:\Users\Ag
19:19:30...	Procmon64.exe	195896	RegOpenKey	HKLM\System\Cur
19:19:30...	Procmon64.exe	195896	RegOpenKey	HKLM\System\Cur
19:19:30...	Procmon64.exe	195896	RegQueryValue	HKLM\System\Cur
19:19:30...	Procmon64.exe	195896	Thread Create	
19:19:30...	Procmon64.exe	195896	RegQueryValue	HKLM\System\Cur
19:19:30...	Procmon64.exe	195896	RegQueryValue	HKLM\System\Cur
19:19:30...	Procmon64.exe	195896	RegQueryValue	HKLM\System\Cur
19:19:30...	Procmon64.exe	195896	RegQueryValue	HKLM\System\Cur
19:19:30...	Procmon64.exe	195896	RegQueryValue	HKLM\System\Cur

```
execve("./hello", ["./hello"], 0x7ffd7bf56360 /* 43 vars */) = 0
brk(NULL)                               = 0x559ccdc05000
arch_prctl(0x3001 /* ARCH_??? */, 0x7ffcc35a8280) = -1 EINVAL (Invalid argument)
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "tls/haswell/x86_64/librt.so.1", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "tls/haswell/librt.so.1", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "tls/x86_64/librt.so.1", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "tls/librt.so.1", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "haswell/x86_64/librt.so.1", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "haswell/librt.so.1", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "x86_64/librt.so.1", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "librt.so.1", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/local/lib/tls/haswell/x86_64/librt.so.1", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
stat("/usr/local/lib/tls/haswell/x86_64", 0x7ffcc35a74d0) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/local/lib/tls/haswell/librt.so.1", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
stat("/usr/local/lib/tls/haswell", 0x7ffcc35a74d0) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/local/lib/tls/x86_64/librt.so.1", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
stat("/usr/local/lib/tls/x86_64", 0x7ffcc35a74d0) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/local/lib/tls/librt.so.1", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
stat("/usr/local/lib/tls", 0x7ffcc35a74d0) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/local/lib/haswell/x86_64/librt.so.1", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
stat("/usr/local/lib/haswell/x86_64", 0x7ffcc35a74d0) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/local/lib/haswell/librt.so.1", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
stat("/usr/local/lib/haswell", 0x7ffcc35a74d0) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/local/lib/x86_64/librt.so.1", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
stat("/usr/local/lib/x86_64", 0x7ffcc35a74d0) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/local/lib/librt.so.1", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
stat("/usr/local/lib", {st_mode=S_IFDIR|0755, st_size=4096, ...}) = 0
openat(AT_FDCWD, "/home/ianni/.local/lib/tls/haswell/x86_64/librt.so.1", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
stat("/home/ianni/.local/lib/tls/haswell/x86_64", 0x7ffcc35a74d0) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/home/ianni/.local/lib/tls/haswell/librt.so.1", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
stat("/home/ianni/.local/lib/tls/haswell", 0x7ffcc35a74d0) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/home/ianni/.local/lib/tls/x86_64/librt.so.1", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
stat("/home/ianni/.local/lib/tls/x86_64", 0x7ffcc35a74d0) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/home/ianni/.local/lib/tls/librt.so.1", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
```

Meet Strace & Procmon

La storia

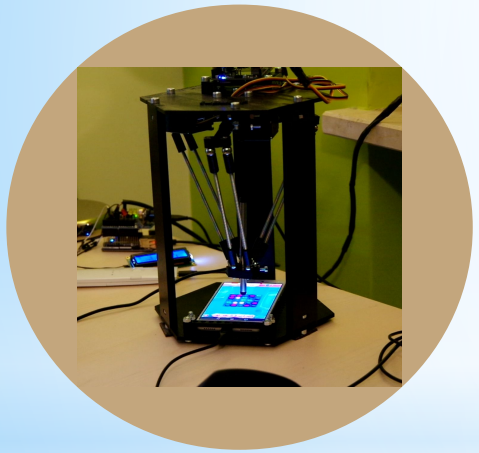

Multics
CP/M
MS-DOS
OS/2
Unix
Windows 3
Windows NT
Linux
FreeBSD, Solaris
Windows 95/98/ME
Windows 2000/XP
Windows Vista, Windows Server 2008
Mac OS
Mac OS X, iOS
Windows 7
Windows 8, 8.1
Ubuntu, Debian,
Mint, Mandriva..
Android, Tizen..
Windows 10
Windows 11


processor
ALU
general registers
instruction decoder
control unit
bus interface
memory
memory array
CS
address decoder
peripheral
peripheral logic
CS
address decoder
address bus
data bus
control bus
DMA controller
HOLD, HLDA
clock
INTR, INTA
interrupt controller

SCAN ME

Elenco completo su
https://sites.google.com/unical.it/iannigbtemitesi/

Temi di tesi e tirocinio



ROBOT

DIGITAL FORENSICS

VIDEOGAME

NETWORK & SECURITY

http://bit.ly/3Y6baEt

https://www.youtube.com/watch?v=pgNjBhVs7_4